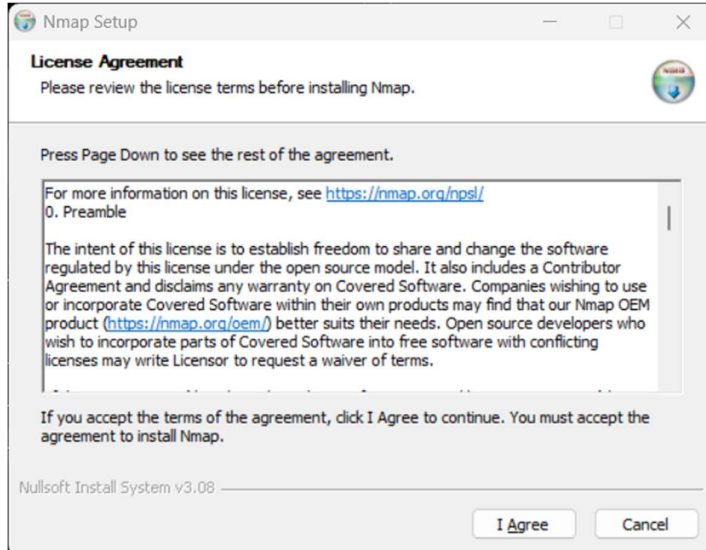


Network Scanning With Nmap

1. Setup of Nmap



2. Basic scan using Nmap (nmap <IP>)

```
PS C:\Users\Akash> nmap 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 01:17 India Standard Time
Nmap scan report for 192.168.1.100
Host is up (0.000042s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
2179/tcp  open  vmrpd
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
```

Basic scan will let you know basic services running and ports in fewer details

3. Service detection (nmap -sV <IP>)

```
PS C:\Users\Akash> nmap -sV 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 01:18 India Standard Time
Nmap scan report for 192.168.1.100
Host is up (0.000061s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
80/tcp    open  http          Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.2.12)
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
443/tcp   open  ssl/http      Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.2.12)
445/tcp   open  microsoft-ds?
2179/tcp  open  vmrpd?
3306/tcp  open  mysql         MariaDB 10.3.23 or earlier (unauthorized)

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.80 seconds
```

Service detection is used when we need in depth details of services and their versions

4. Aggressive scan (nmap -A <IP>)

```
PS C:\Users\Akash> nmap -A 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 01:28 India Standard Time
Nmap scan report for 192.168.1.100
Host is up (0.00066s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.2.12)
|_ http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
|_ http-title: Welcome to XAMPP
|_ Requested resource was http://192.168.1.100/dashboard/
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
443/tcp   open  ssl/http     Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.2.12)
|_ ssl-cert: Subject: commonName=localhost
|_ Not valid before: 2009-11-10T23:48:47
|_ Not valid after: 2019-11-08T23:48:47
|_ http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
|_ ssl-date: TLS randomness does not represent time
|_ tls-alpn:
|_ http/1.1
|_ http-title: Welcome to XAMPP
|_ Requested resource was https://192.168.1.100/dashboard/
445/tcp   open  microsoft-ds?
2179/tcp  open  vmrpd?
3306/tcp  open  mysql        MariaDB 10.3.23 or earlier (unauthorized)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=3/28%OT=80%CT=1%CU=31754%PV=Y%DS=0%DC=L%G=Y%TM=67E5AE0
OS:B%P=i686-pc-windows-windows)SE=IS=108%TI=I%CI=I%II=I%SS=S
OS:%TS=A)SEQ(SP=104%GCD=1%ISR=109%T=I%IT=I%SS=S%TS=A)SEQ(SP=106%GCD=1
OS:%ISR=109%TI=I%CI=I%II=I%SS=S%T=I%IT=I%SS=S%TS=A)SEQ(SP=108%GCD=1%ISR=10A%TI=I%CI=I%II=I
OS:%SS=S%TS=A)OPS(O1=MFFD7NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7
OS:NW8ST11%O5=MFFD7NW8ST11%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF
OS:%W5=FFFF%W6=FFFF%W7=FFFF%W8=FFFF%W9=FFFF%W10=FFFF%W11=FFFF%W12=FFFF%W13=FFFF%W14=FFFF%W15=FFFF%W16=FFFF%W17=FFFF%W18=FFFF%W19=FFFF%W20=FFFF%W21=FFFF%W22=FFFF%W23=FFFF%W24=FFFF%W25=FFFF%W26=FFFF%W27=FFFF%W28=FFFF%W29=FFFF%W30=FFFF%W31=FFFF%W32=FFFF%W33=FFFF%W34=FFFF%W35=FFFF%W36=FFFF%W37=FFFF%W38=FFFF%W39=FFFF%W40=FFFF%W41=FFFF%W42=FFFF%W43=FFFF%W44=FFFF%W45=FFFF%W46=FFFF%W47=FFFF%W48=FFFF%W49=FFFF%W50=FFFF%W51=FFFF%W52=FFFF%W53=FFFF%W54=FFFF%W55=FFFF%W56=FFFF%W57=FFFF%W58=FFFF%W59=FFFF%W60=FFFF%W61=FFFF%W62=FFFF%W63=FFFF%W64=FFFF%W65=FFFF%W66=FFFF%W67=FFFF%W68=FFFF%W69=FFFF%W70=FFFF%W71=FFFF%W72=FFFF%W73=FFFF%W74=FFFF%W75=FFFF%W76=FFFF%W77=FFFF%W78=FFFF%W79=FFFF%W80=FFFF%W81=FFFF%W82=FFFF%W83=FFFF%W84=FFFF%W85=FFFF%W86=FFFF%W87=FFFF%W88=FFFF%W89=FFFF%W90=FFFF%W91=FFFF%W92=FFFF%W93=FFFF%W94=FFFF%W95=FFFF%W96=FFFF%W97=FFFF%W98=FFFF%W99=FFFF%W100=FFFF%W101=FFFF%W102=FFFF%W103=FFFF%W104=FFFF%W105=FFFF%W106=FFFF%W107=FFFF%W108=FFFF%W109=FFFF%W110=FFFF%W111=FFFF%W112=FFFF%W113=FFFF%W114=FFFF%W115=FFFF%W116=FFFF%W117=FFFF%W118=FFFF%W119=FFFF%W120=FFFF%W121=FFFF%W122=FFFF%W123=FFFF%W124=FFFF%W125=FFFF%W126=FFFF%W127=FFFF%W128=FFFF%W129=FFFF%W130=FFFF%W131=FFFF%W132=FFFF%W133=FFFF%W134=FFFF%W135=FFFF%W136=FFFF%W137=FFFF%W138=FFFF%W139=FFFF%W140=FFFF%W141=FFFF%W142=FFFF%W143=FFFF%W144=FFFF%W145=FFFF%W146=FFFF%W147=FFFF%W148=FFFF%W149=FFFF%W150=FFFF%W151=FFFF%W152=FFFF%W153=FFFF%W154=FFFF%W155=FFFF%W156=FFFF%W157=FFFF%W158=FFFF%W159=FFFF%W160=FFFF%W161=FFFF%W162=FFFF%W163=FFFF%W164=FFFF%W165=FFFF%W166=FFFF%W167=FFFF%W168=FFFF%W169=FFFF%W170=FFFF%W171=FFFF%W172=FFFF%W173=FFFF%W174=FFFF%W175=FFFF%W176=FFFF%W177=FFFF%W178=FFFF%W179=FFFF%W180=FFFF%W181=FFFF%W182=FFFF%W183=FFFF%W184=FFFF%W185=FFFF%W186=FFFF%W187=FFFF%W188=FFFF%W189=FFFF%W190=FFFF%W191=FFFF%W192=FFFF%W193=FFFF%W194=FFFF%W195=FFFF%W196=FFFF%W197=FFFF%W198=FFFF%W199=FFFF%W200=FFFF%W201=FFFF%W202=FFFF%W203=FFFF%W204=FFFF%W205=FFFF%W206=FFFF%W207=FFFF%W208=FFFF%W209=FFFF%W210=FFFF%W211=FFFF%W212=FFFF%W213=FFFF%W214=FFFF%W215=FFFF%W216=FFFF%W217=FFFF%W218=FFFF%W219=FFFF%W220=FFFF%W221=FFFF%W222=FFFF%W223=FFFF%W224=FFFF%W225=FFFF%W226=FFFF%W227=FFFF%W228=FFFF%W229=FFFF%W230=FFFF%W231=FFFF%W232=FFFF%W233=FFFF%W234=FFFF%W235=FFFF%W236=FFFF%W237=FFFF%W238=FFFF%W239=FFFF%W240=FFFF%W241=FFFF%W242=FFFF%W243=FFFF%W244=FFFF%W245=FFFF%W246=FFFF%W247=FFFF%W248=FFFF%W249=FFFF%W250=FFFF%W251=FFFF%W252=FFFF%W253=FFFF%W254=FFFF%W255=FFFF%W256=FFFF%W257=FFFF%W258=FFFF%W259=FFFF%W260=FFFF%W261=FFFF%W262=FFFF%W263=FFFF%W264=FFFF%W265=FFFF%W266=FFFF%W267=FFFF%W268=FFFF%W269=FFFF%W270=FFFF%W271=FFFF%W272=FFFF%W273=FFFF%W274=FFFF%W275=FFFF%W276=FFFF%W277=FFFF%W278=FFFF%W279=FFFF%W280=FFFF%W281=FFFF%W282=FFFF%W283=FFFF%W284=FFFF%W285=FFFF%W286=FFFF%W287=FFFF%W288=FFFF%W289=FFFF%W290=FFFF%W291=FFFF%W292=FFFF%W293=FFFF%W294=FFFF%W295=FFFF%W296=FFFF%W297=FFFF%W298=FFFF%W299=FFFF%W300=FFFF%W301=FFFF%W302=FFFF%W303=FFFF%W304=FFFF%W305=FFFF%W306=FFFF%W307=FFFF%W308=FFFF%W309=FFFF%W310=FFFF%W311=FFFF%W312=FFFF%W313=FFFF%W314=FFFF%W315=FFFF%W316=FFFF%W317=FFFF%W318=FFFF%W319=FFFF%W320=FFFF%W321=FFFF%W322=FFFF%W323=FFFF%W324=FFFF%W325=FFFF%W326=FFFF%W327=FFFF%W328=FFFF%W329=FFFF%W330=FFFF%W331=FFFF%W332=FFFF%W333=FFFF%W334=FFFF%W335=FFFF%W336=FFFF%W337=FFFF%W338=FFFF%W339=FFFF%W340=FFFF%W341=FFFF%W342=FFFF%W343=FFFF%W344=FFFF%W345=FFFF%W346=FFFF%W347=FFFF%W348=FFFF%W349=FFFF%W350=FFFF%W351=FFFF%W352=FFFF%W353=FFFF%W354=FFFF%W355=FFFF%W356=FFFF%W357=FFFF%W358=FFFF%W359=FFFF%W360=FFFF%W361=FFFF%W362=FFFF%W363=FFFF%W364=FFFF%W365=FFFF%W366=FFFF%W367=FFFF%W368=FFFF%W369=FFFF%W370=FFFF%W371=FFFF%W372=FFFF%W373=FFFF%W374=FFFF%W375=FFFF%W376=FFFF%W377=FFFF%W378=FFFF%W379=FFFF%W380=FFFF%W381=FFFF%W382=FFFF%W383=FFFF%W384=FFFF%W385=FFFF%W386=FFFF%W387=FFFF%W388=FFFF%W389=FFFF%W390=FFFF%W391=FFFF%W392=FFFF%W393=FFFF%W394=FFFF%W395=FFFF%W396=FFFF%W397=FFFF%W398=FFFF%W399=FFFF%W400=FFFF%W401=FFFF%W402=FFFF%W403=FFFF%W404=FFFF%W405=FFFF%W406=FFFF%W407=FFFF%W408=FFFF%W409=FFFF%W410=FFFF%W411=FFFF%W412=FFFF%W413=FFFF%W414=FFFF%W415=FFFF%W416=FFFF%W417=FFFF%W418=FFFF%W419=FFFF%W420=FFFF%W421=FFFF%W422=FFFF%W423=FFFF%W424=FFFF%W425=FFFF%W426=FFFF%W427=FFFF%W428=FFFF%W429=FFFF%W430=FFFF%W431=FFFF%W432=FFFF%W433=FFFF%W434=FFFF%W435=FFFF%W436=FFFF%W437=FFFF%W438=FFFF%W439=FFFF%W440=FFFF%W441=FFFF%W442=FFFF%W443=FFFF%W444=FFFF%W445=FFFF%W446=FFFF%W447=FFFF%W448=FFFF%W449=FFFF%W450=FFFF%W451=FFFF%W452=FFFF%W453=FFFF%W454=FFFF%W455=FFFF%W456=FFFF%W457=FFFF%W458=FFFF%W459=FFFF%W460=FFFF%W461=FFFF%W462=FFFF%W463=FFFF%W464=FFFF%W465=FFFF%W466=FFFF%W467=FFFF%W468=FFFF%W469=FFFF%W470=FFFF%W471=FFFF%W472=FFFF%W473=FFFF%W474=FFFF%W475=FFFF%W476=FFFF%W477=FFFF%W478=FFFF%W479=FFFF%W480=FFFF%W481=FFFF%W482=FFFF%W483=FFFF%W484=FFFF%W485=FFFF%W486=FFFF%W487=FFFF%W488=FFFF%W489=FFFF%W490=FFFF%W491=FFFF%W492=FFFF%W493=FFFF%W494=FFFF%W495=FFFF%W496=FFFF%W497=FFFF%W498=FFFF%W499=FFFF%W500=FFFF%W501=FFFF%W502=FFFF%W503=FFFF%W504=FFFF%W505=FFFF%W506=FFFF%W507=FFFF%W508=FFFF%W509=FFFF%W510=FFFF%W511=FFFF%W512=FFFF%W513=FFFF%W514=FFFF%W515=FFFF%W516=FFFF%W517=FFFF%W518=FFFF%W519=FFFF%W520=FFFF%W521=FFFF%W522=FFFF%W523=FFFF%W524=FFFF%W525=FFFF%W526=FFFF%W527=FFFF%W528=FFFF%W529=FFFF%W530=FFFF%W531=FFFF%W532=FFFF%W533=FFFF%W534=FFFF%W535=FFFF%W536=FFFF%W537=FFFF%W538=FFFF%W539=FFFF%W540=FFFF%W541=FFFF%W542=FFFF%W543=FFFF%W544=FFFF%W545=FFFF%W546=FFFF%W547=FFFF%W548=FFFF%W549=FFFF%W550=FFFF%W551=FFFF%W552=FFFF%W553=FFFF%W554=FFFF%W555=FFFF%W556=FFFF%W557=FFFF%W558=FFFF%W559=FFFF%W560=FFFF%W561=FFFF%W562=FFFF%W563=FFFF%W564=FFFF%W565=FFFF%W566=FFFF%W567=FFFF%W568=FFFF%W569=FFFF%W570=FFFF%W571=FFFF%W572=FFFF%W573=FFFF%W574=FFFF%W575=FFFF%W576=FFFF%W577=FFFF%W578=FFFF%W579=FFFF%W580=FFFF%W581=FFFF%W582=FFFF%W583=FFFF%W584=FFFF%W585=FFFF%W586=FFFF%W587=FFFF%W588=FFFF%W589=FFFF%W590=FFFF%W591=FFFF%W592=FFFF%W593=FFFF%W594=FFFF%W595=FFFF%W596=FFFF%W597=FFFF%W598=FFFF%W599=FFFF%W600=FFFF%W601=FFFF%W602=FFFF%W603=FFFF%W604=FFFF%W605=FFFF%W606=FFFF%W607=FFFF%W608=FFFF%W609=FFFF%W610=FFFF%W611=FFFF%W612=FFFF%W613=FFFF%W614=FFFF%W615=FFFF%W616=FFFF%W617=FFFF%W618=FFFF%W619=FFFF%W620=FFFF%W621=FFFF%W622=FFFF%W623=FFFF%W624=FFFF%W625=FFFF%W626=FFFF%W627=FFFF%W628=FFFF%W629=FFFF%W630=FFFF%W631=FFFF%W632=FFFF%W633=FFFF%W634=FFFF%W635=FFFF%W636=FFFF%W637=FFFF%W638=FFFF%W639=FFFF%W640=FFFF%W641=FFFF%W642=FFFF%W643=FFFF%W644=FFFF%W645=FFFF%W646=FFFF%W647=FFFF%W648=FFFF%W649=FFFF%W650=FFFF%W651=FFFF%W652=FFFF%W653=FFFF%W654=FFFF%W655=FFFF%W656=FFFF%W657=FFFF%W658=FFFF%W659=FFFF%W660=FFFF%W661=FFFF%W662=FFFF%W663=FFFF%W664=FFFF%W665=FFFF%W666=FFFF%W667=FFFF%W668=FFFF%W669=FFFF%W670=FFFF%W671=FFFF%W672=FFFF%W673=FFFF%W674=FFFF%W675=FFFF%W676=FFFF%W677=FFFF%W678=FFFF%W679=FFFF%W680=FFFF%W681=FFFF%W682=FFFF%W683=FFFF%W684=FFFF%W685=FFFF%W686=FFFF%W687=FFFF%W688=FFFF%W689=FFFF%W690=FFFF%W691=FFFF%W692=FFFF%W693=FFFF%W694=FFFF%W695=FFFF%W696=FFFF%W697=FFFF%W698=FFFF%W699=FFFF%W700=FFFF%W701=FFFF%W702=FFFF%W703=FFFF%W704=FFFF%W705=FFFF%W706=FFFF%W707=FFFF%W708=FFFF%W709=FFFF%W710=FFFF%W711=FFFF%W712=FFFF%W713=FFFF%W714=FFFF%W715=FFFF%W716=FFFF%W717=FFFF%W718=FFFF%W719=FFFF%W720=FFFF%W721=FFFF%W722=FFFF%W723=FFFF%W724=FFFF%W725=FFFF%W726=FFFF%W727=FFFF%W728=FFFF%W729=FFFF%W730=FFFF%W731=FFFF%W732=FFFF%W733=FFFF%W734=FFFF%W735=FFFF%W736=FFFF%W737=FFFF%W738=FFFF%W739=FFFF%W740=FFFF%W741=FFFF%W742=FFFF%W743=FFFF%W744=FFFF%W745=FFFF%W746=FFFF%W747=FFFF%W748=FFFF%W749=FFFF%W750=FFFF%W751=FFFF%W752=FFFF%W753=FFFF%W754=FFFF%W755=FFFF%W756=FFFF%W757=FFFF%W758=FFFF%W759=FFFF%W760=FFFF%W761=FFFF%W762=FFFF%W763=FFFF%W764=FFFF%W765=FFFF%W766=FFFF%W767=FFFF%W768=FFFF%W769=FFFF%W770=FFFF%W771=FFFF%W772=FFFF%W773=FFFF%W774=FFFF%W775=FFFF%W776=FFFF%W777=FFFF%W778=FFFF%W779=FFFF%W780=FFFF%W781=FFFF%W782=FFFF%W783=FFFF%W784=FFFF%W785=FFFF%W786=FFFF%W787=FFFF%W788=FFFF%W789=FFFF%W790=FFFF%W791=FFFF%W792=FFFF%W793=FFFF%W794=FFFF%W795=FFFF%W796=FFFF%W797=FFFF%W798=FFFF%W799=FFFF%W800=FFFF%W801=FFFF%W802=FFFF%W803=FFFF%W804=FFFF%W805=FFFF%W806=FFFF%W807=FFFF%W808=FFFF%W809=FFFF%W810=FFFF%W811=FFFF%W812=FFFF%W813=FFFF%W814=FFFF%W815=FFFF%W816=FFFF%W817=FFFF%W818=FFFF%W819=FFFF%W820=FFFF%W821=FFFF%W822=FFFF%W823=FFFF%W824=FFFF%W825=FFFF%W826=FFFF%W827=FFFF%W828=FFFF%W829=FFFF%W830=FFFF%W831=FFFF%W832=FFFF%W833=FFFF%W834=FFFF%W835=FFFF%W836=FFFF%W837=FFFF%W838=FFFF%W839=FFFF%W840=FFFF%W841=FFFF%W842=FFFF%W843=FFFF%W844=FFFF%W845=FFFF%W846=FFFF%W847=FFFF%W848=FFFF%W849=FFFF%W850=FFFF%W851=FFFF%W852=FFFF%W853=FFFF%W854=FFFF%W855=FFFF%W856=FFFF%W857=FFFF%W858=FFFF%W859=FFFF%W860=FFFF%W861=FFFF%W862=FFFF%W863=FFFF%W864=FFFF%W865=FFFF%W866=FFFF%W867=FFFF%W868=FFFF%W869=FFFF%W870=FFFF%W871=FFFF%W872=FFFF%W873=FFFF%W874=FFFF%W875=FFFF%W876=FFFF%W877=FFFF%W878=FFFF%W879=FFFF%W880=FFFF%W881=FFFF%W882=FFFF%W883=FFFF%W884=FFFF%W885=FFFF%W886=FFFF%W887=FFFF%W888=FFFF%W889=FFFF%W890=FFFF%W891=FFFF%W892=FFFF%W893=FFFF%W894=FFFF%W895=FFFF%W896=FFFF%W897=FFFF%W898=FFFF%W899=FFFF%W900=FFFF%W901=FFFF%W902=FFFF%W903=FFFF%W904=FFFF%W905=FFFF%W906=FFFF%W907=FFFF%W908=FFFF%W909=FFFF%W910=FFFF%W911=FFFF%W912=FFFF%W913=FFFF%W914=FFFF%W915=FFFF%W916=FFFF%W917=FFFF%W918=FFFF%W919=FFFF%W920=FFFF%W921=FFFF%W922=FFFF%W923=FFFF%W924=FFFF%W925=FFFF%W926=FFFF%W927=FFFF%W928=FFFF%W929=FFFF%W930=FFFF%W931=FFFF%W932=FFFF%W933=FFFF%W934=FFFF%W935=FFFF%W936=FFFF%W937=FFFF%W938=FFFF%W939=FFFF%W940=FFFF%W941=FFFF%W942=FFFF%W943=FFFF%W944=FFFF%W945=FFFF%W946=FFFF%W947=FFFF%W948=FFFF%W949=FFFF%W950=FFFF%W951=FFFF%W952=FFFF%W953=FFFF%W954=FFFF%W955=FFFF%W956=FFFF%W957=FFFF%W958=FFFF%W959=FFFF%W960=FFFF%W961=FFFF%W962=FFFF%W963=FFFF%W964=FFFF%W965=FFFF%W966=FFFF%W967=FFFF%W968=FFFF%W969=FFFF%W970=FFFF%W971=FFFF%W972=FFFF%W973=FFFF%W974=FFFF%W975=FFFF%W976=FFFF%W977=FFFF%W978=FFFF%W979=FFFF%W980=FFFF%W981=FFFF%W982=FFFF%W983=FFFF%W984=FFFF%W985=FFFF%W986=FFFF%W987=FFFF%W988=FFFF%W989=FFFF%W990=FFFF%W991=FFFF%W992=FFFF%W993=FFFF%W994=FFFF%W995=FFFF%W996=FFFF%W997=FFFF%W998=FFFF%W999=FFFF%W1000=FFFF%W1001=FFFF%W1002=FFFF%W1003=FFFF%W1004=FFFF%W1005=FFFF%W1006=FFFF%W1007=FFFF%W1008=FFFF%W1009=FFFF%W1010=FFFF%W1011=FFFF%W1012=FFFF%W10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